SKILL WEEK-2

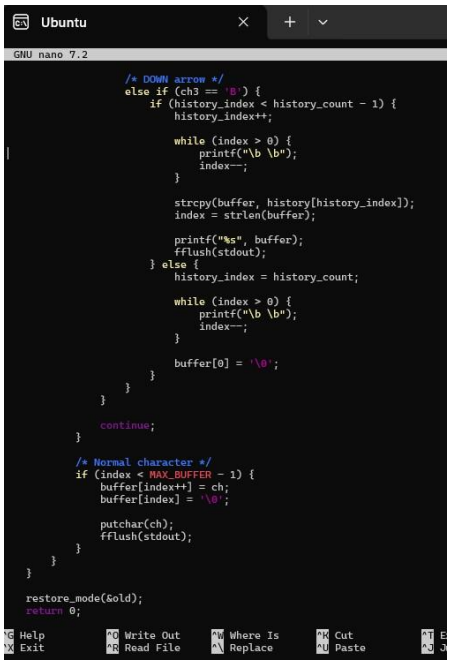
Session3:

Task1: To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality.

Source code:

```
  GNU nano 7.2
#include <stdio.h>
#include <string.h>
#include <termios.h>
#include <unistd.h>

#define MAX_BUFFER 100
#define MAX_HISTORY 10

char history[MAX_HISTORY][MAX_BUFFER];
int history_count = 0;
int history_index = 0;

void enable_raw_mode(struct termios *old) {
    struct termios raw;

    tcgetattr(STDIN_FILENO, old);
    raw = *old;

    raw.c_lflag &= ~(ICANON | ECHO);
    tcsetattr(STDIN_FILENO, TCSAFLUSH, &raw);
}

void restore_mode(struct termios *old) {
    tcsetattr(STDIN_FILENO, TCSAFLUSH, old);
}

void save_command(char *command) {
    if (strlen(command) == 0)
        return;

    if (history_count < MAX_HISTORY) {
        strcpy(history[history_count], command);
        history_count++;
    } else {
        for (int i = 1; i < MAX_HISTORY; i++)
            strcpy(history[i - 1], history[i]);

        strcpy(history[MAX_HISTORY - 1], command);
    }

    history_index = history_count;
}
```

```
Ubuntu
  GNU nano 7.2
int main() {
    struct termios old;
    char buffer[MAX_BUFFER];
    int index = 0;
    int ch;

    printf("Session 3 - Command History\n");
    printf("Type commands. Use UP/DOWN arrows for history.\n");
    printf("Press Ctrl+C to stop.\n\n");

    enable_raw_mode(&old);

    while (1) {
        index = 0;
        buffer[0] = '\0';

        printf("myshell> ");
        fflush(stdout);

        while (1) {
            ch = getchar();

            /* Enter key */
            if (ch == '\n' || ch == '\r') {
                buffer[index] = '\0';

                printf("\n");

                if (strcmp(buffer, "exit") == 0) {
                    restore_mode(&old);
                    printf("Exiting...\n");
                    return 0;
                }

                if (strlen(buffer) > 0) {
                    printf("Command: %s\n", buffer);
                    save_command(buffer);
                }

                break;
 |          }
^G Help        ^O Write Out   ^W Where Is    ^K Cut         ^T Execu
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justi
```



Output:

```
avulasanjana@SanjanaReddy:~$ nano session3_task1.c
avulasanjana@SanjanaReddy:~$ ^C
avulasanjana@SanjanaReddy:~$ gcc session3_task1.c -o session3_task1
avulasanjana@SanjanaReddy:~$ ./session3_task1
Session 3 - Command History
Type commands. Use UP/DOWN arrows for history.
Press Ctrl+C to stop.

myshell> hello
You entered: hello
myshell> test
You entered: test
```

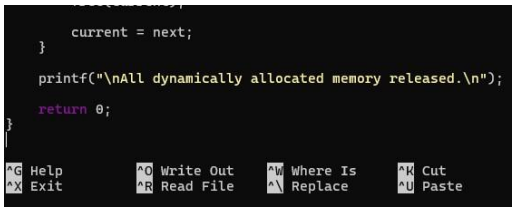
Observation: I learned how to use escape sequences to detect special keyboard inputs such as the Up and Down arrow keys. I learned how to store previously entered commands in a history buffer and navigate through them using the arrow keys. I also learned how to update the input buffer when recalling previous commands and manage user input in an interactive shell.

Task2: To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

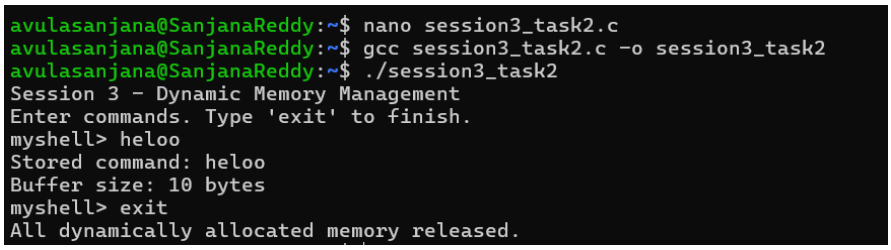
Source code:


GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#define INITIAL_SIZE 10
typedef struct Node {
char *command;
struct Node *next;
} Node;
int main() {
char *buffer;
size_t size = INITIAL_SIZE;
size_t length = 0;
int ch;
Node *head = NULL;
Node *tail = NULL;
buffer = malloc(size);
if (buffer == NULL) {
perror("malloc failed");
return 1;
}
printf("Session 3 - Dynamic Memory Management\n");
printf("Enter commands. Type 'exit' to finish.\n\n");
while (1) {
length = 0;
printf("myshell> ");
fflush(stdout);
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste


GNU nano 7.2
while ((ch = getchar()) != '\n' && ch != EOF) {
if (length + 1 >= size) {
size *= 2;
char *temp = realloc(buffer, size);
if (temp == NULL) {
perror("realloc failed");
free(buffer);
return 1;
}
buffer = temp;
}
buffer[length++] = ch;
}
buffer[length] = '\0';
if (strcmp(buffer, "exit") == 0) {
break;
}
if (length == 0) {
continue;
}
Node *new_node = malloc(sizeof(Node));
if (new_node == NULL) {
perror("malloc failed");
break;
}
^G Help ^O Write Out ^W Where Is ^K Cut
^X Exit ^R Read File ^\ Replace ^U Paste



Output:



Observation: I learned how to use dynamic memory allocation in C using malloc() and realloc(). I learned how to resize a buffer when more memory is required and how to store commands using a linked list. I also learned the importance of releasing dynamically allocated memory using free() to prevent memory leaks.

Session4

Task1: To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

Source code:

Output:

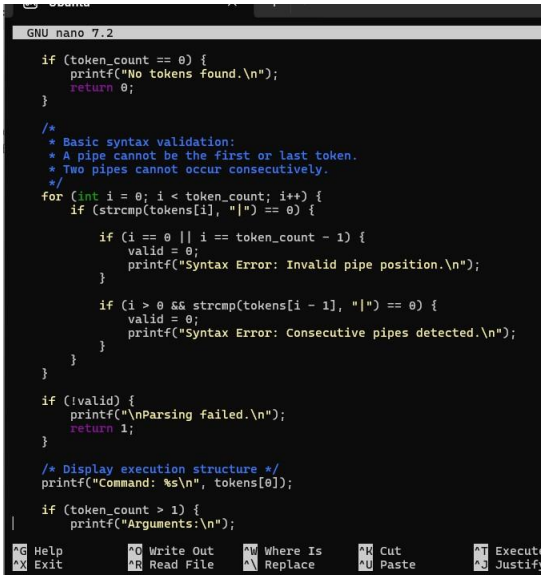
```
avulasanjana@SanjanaReddy:~$ nano session4_task1.c
avulasanjana@SanjanaReddy:~$ gcc session4_task1.c -o session4_task1
avulasanjana@SanjanaReddy:~$ ./session4_task1
Session 4 - Tokenization
Enter a command: ls -l | grep txt

Tokens identified:
Token 1: [ls]
Token 2: [-l]
Token 3: [grep]
Token 4: [txt]

Token stream validation:
Token stream is valid.
avulasanjana@SanjanaReddy:~$
```

Observation: I learned how to split a command into individual tokens using tokenization. I understood how whitespace and special delimiters such as | can separate different parts of a command. I also learned how to store tokens in a token structure and validate the resulting token stream before further processing.

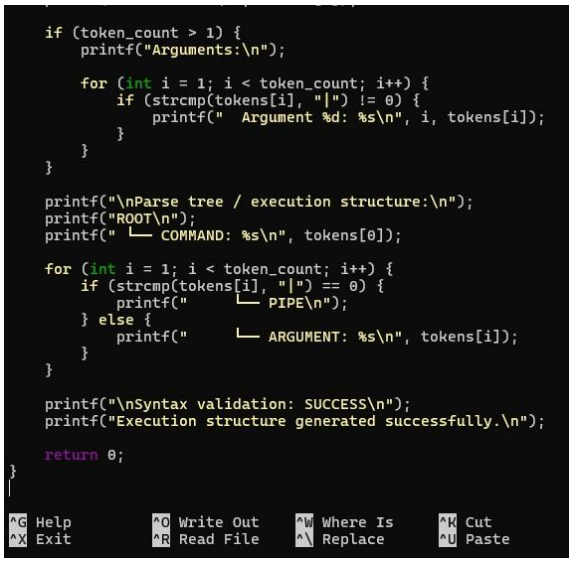
Task2: To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

Source code:

```
 GNU nano 7.2
#include <stdio.h>
#include <string.h>

#define MAX_TOKENS 50

int main() {
    char input[500];
    char *tokens[MAX_TOKENS];
    int token_count = 0;
    int valid = 1;

    printf("Session 4 - Parser\n");
    printf("Enter a command: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    /* Handle empty command */
    if (strlen(input) == 0) {
        printf("Empty command detected.\n");
        return 0;
    }

    /* Tokenize input */
    char *token = strtok(input, " \t");

    while (token != NULL && token_count < MAX_TOKENS) {
        tokens[token_count++] = token;
        token = strtok(NULL, " \t");
    }

    printf("\nParse Structure:\n");

    if (token_count == 0) {
        printf("No tokens found.\n");
        return 0;

^G Help         ^O Write Out    ^W Where Is     ^K Cut
^X Exit         ^R Read File    ^\ Replace      ^U Paste
```



```
GNU nano 7.2

    if (token_count == 0) {
        printf("No tokens found.\n");
        return 0;
    }

    /*
     * Basic syntax validation:
     * A pipe cannot be the first or last token.
     * Two pipes cannot occur consecutively.
     */
    for (int i = 0; i < token_count; i++) {
        if (strcmp(tokens[i], "|") == 0) {

            if (i == 0 || i == token_count - 1) {
                valid = 0;
                printf("Syntax Error: Invalid pipe position.\n");
            }

            if (i > 0 && strcmp(tokens[i - 1], "|") == 0) {
                valid = 0;
                printf("Syntax Error: Consecutive pipes detected.\n");
            }
        }
    }

    if (!valid) {
        printf("\nParsing failed.\n");
        return 1;
    }

    /* Display execution structure */
    printf("Command: %s\n", tokens[0]);

    if (token_count > 1) {
        printf("Arguments:\n");

^G Help         ^O Write Out    ^W Where Is     ^K Cut          ^T Execut
^X Exit         ^R Read File    ^\ Replace      ^U Paste        ^J Justif
```



```
    if (token_count > 1) {
        printf("Arguments:\n");

        for (int i = 1; i < token_count; i++) {
            if (strcmp(tokens[i], "|") != 0) {
                printf("  Argument %d: %s\n", i, tokens[i]);
            }
        }
    }

    printf("\nParse tree / execution structure:\n");
    printf("ROOT\n");
    printf(" └── COMMAND: %s\n", tokens[0]);

    for (int i = 1; i < token_count; i++) {
        if (strcmp(tokens[i], "|") == 0) {
            printf("      └── PIPE\n");
        } else {
            printf("      └── ARGUMENT: %s\n", tokens[i]);
        }
    }

    printf("\nSyntax validation: SUCCESS\n");
    printf("Execution structure generated successfully.\n");

    return 0;
}

^G Help         ^O Write Out    ^W Where Is     ^K Cut
^X Exit         ^R Read File    ^\ Replace      ^U Paste
```

Output:

```
avulasanjana@SanjanaReddy:~$ nano session4_task2.c
avulasanjana@SanjanaReddy:~$ gcc session4_task2.c -o session4_task2
avulasanjana@SanjanaReddy:~$ ./session4_task2
Session 4 - Parser
Enter a command: ls -l | grep txt

Parse Structure:
Command: ls

Command: ls
Arguments:
  Argument 1: -l
  Argument 3: grep
  Argument 4: txt

Parse tree / execution structure:
ROOT
 └── COMMAND: ls
      └── ARGUMENT: -l
      └── PIPE
      └── ARGUMENT: grep
      └── ARGUMENT: txt

Syntax validation: SUCCESS
Execution structure generated successfully.
avulasanjana@SanjanaReddy:~$
```

Observation: I learned how to design basic parser logic for processing user commands. I understood how to validate command syntax, detect errors such as invalid or consecutive pipe symbols, and handle empty commands. I also learned how tokens can be organized into a parse structure containing commands, arguments, and pipes, which can later be used for command execution.